Personal AI Collaboration Across Physical and Digital Activities

Research interests and preliminary PhD directions

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Research motivation

I want to investigate personal AI collaborators that people can converse with while doing things: assistants that connect dialogue with the surrounding environment, help reason about alternatives, carry decisions into useful actions, and learn from repeated collaboration. Their presence could be a voice, a spatial model, information beside an object, or a completed digital operation. My longer-term interest is how shared experience makes such collaboration coherent and personally meaningful.

My architectural education and work on immersive installations motivate this interest. In spatial work, people move between observing a physical setting, discussing possibilities and changing a digital representation. An assistant must follow that evolving activity and let the person inspect and revise its understanding. I would like to study this collaboration through a bounded, implemented system, with a first activity selected and refined through observation.

An illustrative collaboration

Consider planning a small installation. A person looks around a room and discusses how visitors should move through it. The assistant relates references such as "this panel" and "the view from the entrance" to a known digital model, compares possible layouts, and previews a selected change. The person can point out a mistaken object, demonstrate a different orientation or revise the visitor route. Later, they return with a new requirement and decide together which earlier choices remain useful. This is a proposed test scenario, not an implemented or evaluated capability.

Connections to the lab

TOM provides a platform for context-aware wearable assistance and proof-of-concept activity services [1]. CRAFT connects real-world encounters to an evolving creative artefact and maintains creative context across sessions [2]. PersonaMail supports intent articulation, granular revision and reuse of context-sensitive preferences [3]. These projects connect to my interest in situated collaboration, useful action and adaptation through use.

Existing spatial authoring and context-aware task assistants also provide important precedents [4, 5]. Shared context, editable output, personalisation and memory are therefore foundations to build on. My initial task is to establish a specific interaction problem and test whether an additional design improves collaboration over a capable existing assistant.

Candidate research questions

RQ1: When people move between observing a physical setting and editing a digital artefact, how do they establish and repair shared understanding of objects, goals and constraints with an AI assistant?

RQ2: What interaction design helps people turn a discussion of alternatives into inspectable and revisable actions, and how does it affect correction effort and the quality of the resulting work?

RQ3: Across repeated collaboration, which decisions and preferences should carry forward, and how should they be reconsidered when the activity changes?

An initial research path

I would begin by observing one recurring design or making activity with an evolving artefact. Spatial design or installation planning is a candidate grounded in my experience. Observation and replay interviews would examine how people refer to objects, explain desired effects, compare alternatives and repair an earlier interpretation. Unchanged work, changed requirements and assistant misunderstandings should be distinguished. This phase would determine the first interaction problem and whether wearable sensing supplies information that ordinary workspace records cannot provide.

A first prototype could combine speech and pointing with a known digital scene and a small set of operations such as selecting, moving, rotating and comparing objects. Existing speech and language-model components would support interpretation. I would explore a shared activity view linking relevant objects, current decisions and proposed next actions. Users could inspect alternatives, revise a referent or goal, and confirm reversible changes. Physical rearrangement would remain a human action. The form and value of this view would be tested rather than assumed.

The principal comparison would be a capable conversational assistant with the same model, scene information, history, action tools, previews, confirmation and undo. It would also be able to ask questions and accept corrections. The added activity view and its revision workflow would initially be evaluated as a complete interaction design; a separate controlled comparison would be needed to attribute an effect to the representation alone.

Evaluation would examine misunderstanding repair, actions that violate current requirements, valid work preserved, time to useful progress, and independently assessable artefact quality. Workload, perceived control and collaboration experience would complement these outcomes. Repeated sessions could then examine the reuse and revision of decisions or preferences. Sample size and duration would follow formative observations, pilot variability and the primary outcome; no participant study of this proposal has yet been conducted.

Preparation and scope

Game Fellows provides experience with a rule-based Pygame collaborator, action-linked cues and controlled study preparation. My CAVE internship supports spatial and system-integration work; my capstone contributes reproducible evaluation and error analysis. Wearable integration and real-time multimodal interaction are technical skills I would develop. Early prototyping can use desktop input and known scene data before testing whether a wearable configuration offers additional value.

The intended outcome is evidence about a specific collaboration mechanism, supported by a working prototype and an evaluation that states its task boundaries. Long-term relationship formation remains a broader motivation requiring a separate definition and longer observation. Ethics review would precede recruitment; recording and external model processing would be disclosed, limited to the task, and under participant control.

References

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